



April 17, 2025

VitalConnect, Inc.
Gabriel Nallathambi
Director, Algorithm and Data Science
2870 Zanker Road
Suite 100
San Jose, California 95134

Re: K242129
Trade/Device Name: VitalRhythm
Regulation Number: 21 CFR 870.1425
Regulation Name: Programmable Diagnostic Computer
Regulatory Class: Class II
Product Code: DQK, DPS, QYX
Dated: March 18, 2025
Received: March 19, 2025

Dear Gabriel Nallathambi:

We have reviewed your section 510(k) premarket notification of intent to market the device referenced above and have determined the device is substantially equivalent (for the indications for use stated in the enclosure) to legally marketed predicate devices marketed in interstate commerce prior to May 28, 1976, the enactment date of the Medical Device Amendments, or to devices that have been reclassified in accordance with the provisions of the Federal Food, Drug, and Cosmetic Act (the Act) that do not require approval of a premarket approval application (PMA). You may, therefore, market the device, subject to the general controls provisions of the Act. Although this letter refers to your product as a device, please be aware that some cleared products may instead be combination products. The 510(k) Premarket Notification Database available at <https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpmn/pmn.cfm> identifies combination product submissions. The general controls provisions of the Act include requirements for annual registration, listing of devices, good manufacturing practice, labeling, and prohibitions against misbranding and adulteration. Please note: CDRH does not evaluate information related to contract liability warranties. We remind you, however, that device labeling must be truthful and not misleading.

If your device is classified (see above) into either class II (Special Controls) or class III (PMA), it may be subject to additional controls. Existing major regulations affecting your device can be found in the Code of Federal Regulations, Title 21, Parts 800 to 898. In addition, FDA may publish further announcements concerning your device in the Federal Register.

Additional information about changes that may require a new premarket notification are provided in the FDA guidance documents entitled "Deciding When to Submit a 510(k) for a Change to an Existing Device" (<https://www.fda.gov/media/99812/download>) and "Deciding When to Submit a 510(k) for a Software Change to an Existing Device" (<https://www.fda.gov/media/99785/download>).

Your device is also subject to, among other requirements, the Quality System (QS) regulation (21 CFR Part 820), which includes, but is not limited to, 21 CFR 820.30, Design controls; 21 CFR 820.90, Nonconforming product; and 21 CFR 820.100, Corrective and preventive action. Please note that regardless of whether a change requires premarket review, the QS regulation requires device manufacturers to review and approve changes to device design and production (21 CFR 820.30 and 21 CFR 820.70) and document changes and approvals in the device master record (21 CFR 820.181).

Please be advised that FDA's issuance of a substantial equivalence determination does not mean that FDA has made a determination that your device complies with other requirements of the Act or any Federal statutes and regulations administered by other Federal agencies. You must comply with all the Act's requirements, including, but not limited to: registration and listing (21 CFR Part 807); labeling (21 CFR Part 801); medical device reporting (reporting of medical device-related adverse events) (21 CFR Part 803) for devices or postmarketing safety reporting (21 CFR Part 4, Subpart B) for combination products (see <https://www.fda.gov/combination-products/guidance-regulatory-information/postmarketing-safety-reporting-combination-products>); good manufacturing practice requirements as set forth in the quality systems (QS) regulation (21 CFR Part 820) for devices or current good manufacturing practices (21 CFR Part 4, Subpart A) for combination products; and, if applicable, the electronic product radiation control provisions (Sections 531-542 of the Act); 21 CFR Parts 1000-1050.

All medical devices, including Class I and unclassified devices and combination product device constituent parts are required to be in compliance with the final Unique Device Identification System rule ("UDI Rule"). The UDI Rule requires, among other things, that a device bear a unique device identifier (UDI) on its label and package (21 CFR 801.20(a)) unless an exception or alternative applies (21 CFR 801.20(b)) and that the dates on the device label be formatted in accordance with 21 CFR 801.18. The UDI Rule (21 CFR 830.300(a) and 830.320(b)) also requires that certain information be submitted to the Global Unique Device Identification Database (GUDID) (21 CFR Part 830 Subpart E). For additional information on these requirements, please see the UDI System webpage at <https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance/unique-device-identification-system-udi-system>.

Also, please note the regulation entitled, "Misbranding by reference to premarket notification" (21 CFR 807.97). For questions regarding the reporting of adverse events under the MDR regulation (21 CFR Part 803), please go to <https://www.fda.gov/medical-devices/medical-device-safety/medical-device-reporting-mdr-how-report-medical-device-problems>.

For comprehensive regulatory information about medical devices and radiation-emitting products, including information about labeling regulations, please see Device Advice (<https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance>) and CDRH Learn (<https://www.fda.gov/training-and-continuing-education/cdrh-learn>). Additionally, you may contact the Division of Industry and Consumer Education (DICE) to ask a question about a specific regulatory topic. See the DICE website (<https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory->

[assistance/contact-us-division-industry-and-consumer-education-dice](#)) for more information or contact DICE by email (DICE@fda.hhs.gov) or phone (1-800-638-2041 or 301-796-7100).

Sincerely,

Jennifer W. Shih -S

Jennifer Kozen
Assistant Director
Division of Cardiac Electrophysiology,
Diagnostics, and Monitoring Devices
Office of Cardiovascular Devices
Office of Product Evaluation and Quality
Center for Devices and Radiological Health

Enclosure

Indications for Use

510(k) Number (if known)

K242129

Device Name

VitalRhythm

Indications for Use (Describe)

VitalRhythm is a cloud-based software application for continuous and automatic analysis of cardiac arrhythmias. VitalRhythm is compatible with the "Vista Solution" platform, which includes the VitalPatch biosensor and VistaCenter web application. VitalRhythm is intended to be used for outpatient cardiac telemetry and patient monitoring in non-critical healthcare settings for non-urgent clinical decision-making. VitalRhythm provides analysis of cardiac arrhythmias using ECG data and RR-interval from the VitalPatch in patients who are 18 years of age or older. Results of the VitalRhythm are displayed within the VistaCenter web application to be reviewed and confirmed by qualified healthcare professionals and/or cardiac technicians. VitalRhythm is not intended for use in life-supporting or sustaining systems or for critical care monitoring. The arrhythmia analysis results are not intended to be the sole means of diagnosis and are offered on an advisory basis only, in conjunction with the physician's knowledge of ECG patterns, patient background, clinical history, symptoms, and other diagnostic information.

Type of Use (Select one or both, as applicable)

☒ Prescription Use (Part 21 CFR 801 Subpart D)

☐ Over-The-Counter Use (21 CFR 801 Subpart C)

CONTINUE ON A SEPARATE PAGE IF NEEDED.

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VitalConnect, Inc.
Traditional 510(k): 510(k) Summary

VitalRhythm Arrhythmia Detection Software

510(k) Notification K242129

I. General Information

Applicant Name and Address:

VitalConnect, Inc.
2870 Zanker Road, Suite 100
San Jose, CA 95134 USA
Phone: 408-963-4600

Applicant Contact Person:

Dr. Gabriel Nallathambi
Director, Algorithm and Data Science
Phone: 408-963-4600
Email: gnallathambi@vitalconnect.com

Date Prepared: April 15, 2025

II. Device Information

Device Trade name:

VitalRhythm

Common Name:

Programmable Diagnostic Computer

Classification Name:

Programmable Diagnostic Computer
Electrocardiograph
Arrhythmia detector and alarm (including ST-segment measurement and alarm)

Regulation Numbers:

21CFR§870.1425
21CFR§870.2340
21CFR§870.1025

Product Codes:

DQK - Computer, Diagnostic, Programmable
DPS – Electrocardiograph
QYX – Outpatient Cardiac Telemetry

III. Legally Marketed Predicate Device

Predicate Device: Medicalgorithmics S.A. DeepRhythmAI (K241197)

Reference Device: B-Secur Limited HeartKey® Rhythm (K233755)

Reference Device: VitalConnect, Inc. Vista Solution (K193343)

Reference Device: VitalConnect, Inc. VitalPatch Biosensor (K192757)

IV. Device Description Summary

VitalRhythm is a cloud-based, arrhythmia detection software application that is compatible and intended to be used with the VitalConnect “Vista Solution” platform. The Vista Solution Platform consists of the VitalPatch biosensor, a phone (VistaPhone) or tablet (VistaTablet) relay device preloaded with the VistaPoint software application, VC Cloud and the cloud-based VistaCenter user interface.

VitalRhythm is intended to be used for outpatient cardiac telemetry and patient monitoring in non-critical healthcare settings for non-urgent clinical decision-making.

VitalRhythm analyzes ECG data and RR-interval from the VitalPatch biosensor for reporting of cardiac arrhythmias to be reviewed and adjudicated by qualified healthcare professionals and/or cardiac technicians.

The cloud-based VitalRhythm software application supports the analysis of ECG data and RR-interval using a proprietary algorithm developed using deep learning techniques. The application works in the following way:

1. VitalRhythm accepts ECG and RR-interval data transmitted from the VitalPatch via a secure, cloud-based API (Application Programming Interface)
2. ECG and RR-interval data are analyzed by VitalRhythm using a proprietary algorithm, which detects the following cardiac rhythms:
 - Atrial fibrillation/atrial flutter
 - AV Block (2nd degree, Type I and II)
 - Pause
 - Paroxysmal supraventricular tachycardia (PSVT)
 - Ventricular tachycardia/run
 - Sinus bradycardia
 - Sinus tachycardia
 - Normal sinus rhythm
 - Others (inconclusive)
3. For use with Mobile Cardiac Telemetry (MCT), i.e., outpatient cardiac telemetry: any of the above listed arrhythmias that are detected by the software algorithm are displayed in VistaCenter to be reviewed and analyzed by a qualified cardiac technician in the 24/7

attended Cardiac Monitoring Center, prior to transmitting a notifiable event consistent with the prescribed notification criteria to the prescribing physician during the monitoring period. An event report is generated by VistaCenter as a result of the analysis.

For use with patient monitoring in non-critical healthcare settings: any of the above listed arrhythmias that are detected by the software algorithm are displayed and notified in VistaCenter, in accordance with the notification criteria, during the monitoring period to be reviewed by the prescribing healthcare professional for non-urgent clinical decision-making.

The features, operating procedures, and mitigations in place for the compatible devices ensure continuous data collection and transmission to support the VitalRhythm application for the intended use. Further information is provided in the VitalRhythm *Instructions for Use* document.

V. Intended Use / Indications for Use

VitalRhythm is a cloud-based software application for continuous and automatic analysis of cardiac arrhythmias. VitalRhythm is compatible with the “Vista Solution” platform, which includes the VitalPatch biosensor and VistaCenter web application. VitalRhythm is intended to be used for outpatient cardiac telemetry and patient monitoring in non-critical healthcare settings for non-urgent clinical decision-making. VitalRhythm provides analysis of cardiac arrhythmias using ECG data and RR-interval from the VitalPatch in patients who are 18 years of age or older. Results of the VitalRhythm are displayed within the VistaCenter web application to be reviewed and confirmed by qualified healthcare professionals and/or cardiac technicians. VitalRhythm is not intended for use in life-supporting or sustaining systems or for critical care monitoring. The arrhythmia analysis results are not intended to be the sole means of diagnosis and are offered on an advisory basis only, in conjunction with the physician’s knowledge of ECG patterns, patient background, clinical history, symptoms, and other diagnostic information.

VI. Comparison to Predicate Device

The following table provides a comparison of the intended use and device characteristics of the proposed VitalRhythm device, the DeepRhythmAI predicate device, the HeartKey Rhythm reference device and the compatible VitalConnect reference devices.

VitalRhythm (VitalConnect, Inc.) K242129	DeepRhythmAI (Medicalgorithmics S.A.) K241197	HeartKey® Rhythm (B-Secur Limited) K233755	Vista Solution (VitalConnect, Inc.) K193343	VitalPatch Biosensor (VitalConnect, Inc.) K192757
<i>Subject Device</i>	<i>Predicate Device</i>	<i>Reference Device</i>	<i>Reference Device</i>	<i>Reference Device</i>

Indications for Use	VitalRhythm is a cloud-based software application for continuous and automatic analysis of cardiac arrhythmias. VitalRhythm is compatible with the “Vista Solution” platform, which includes the VitalPatch biosensor and VistaCenter web application. VitalRhythm is intended to be used for outpatient cardiac telemetry and patient monitoring in non-critical healthcare settings for non-urgent clinical decision-making. VitalRhythm provides analysis of cardiac arrhythmias using ECG data and RR-interval from the VitalPatch in patients who are 18 years of age or older. Results of the VitalRhythm are displayed within the VistaCenter web application to be reviewed and confirmed by qualified healthcare professionals and/or cardiac technicians. VitalRhythm is not intended for use in	DeepRhythmAI is a cloud-based software for the assessment of cardiac arrhythmias using two lead ECG data in adult patients. It is intended for use by a healthcare solution integrator to build web, mobile or another types of applications to let qualified healthcare professionals review and confirm the analytic result. The product supports downloading and analyzing data recorded in the compatible formats from dedicated ambulatory ECG devices such as Holter, event recorder, Outpatient Cardiac Telemetry devices or other similar recorders when the assessment of the rhythm is necessary. The product can be electronically interfaced and perform analysis with data transferred from other computer-based ECG systems, such as an ECG management system. DeepRhythmAI	HeartKey Rhythm is intended for use by qualified health-care professionals for analyzing ECG data from adults. HeartKey Rhythm analyzes a single ECG lead recorded from compatible devices used for arrhythmia diagnostics such as Holters, Extended Holters, ECG Patches, Event Recorders, or other similar devices when assessment of the rhythm is necessary. The HeartKey Rhythm application program interface (API) is intended to be integrated into other device software to provide the following capabilities: • ECG signal conditioning, • ECG signal quality assessment, • QRS beat detection, • Ventricular ectopic beat detection, • ECG rhythm analysis HeartKey Rhythm is not intended for use in life supporting, or sustaining systems, or real-time ECG	Vista Solution 2.1 is a software user interface intended for use by healthcare professionals to display physiological data collected by the VitalConnect Platform wireless remote monitoring system in home and healthcare settings. Vista Solution is a secondary, adjunct patient monitor and is not intended to replace existing standard-of-care patient monitoring practices.	The VitalConnect Platform is a wireless remote monitoring system intended for use by healthcare professionals for continuous collection of physiological data in home and healthcare settings. This can include heart rate, electro-cardiography (ECG), heart rate variability, R-R interval, respiratory rate, body temperature, skin temperature, activity (including step count), and posture (body position relative to gravity including fall). Data are transmitted wirelessly from the VitalConnect Sensor for storage and analysis. The VitalConnect Platform can include the ability to notify healthcare professionals when physiological data fall outside selected parameters. The device is intended for use on general care patients who are 18 years of age or older as a general patient monitor, to provide physiological information. The
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	life-supporting or sustaining systems or for critical care monitoring. The arrhythmia analysis results are not intended to be the sole means of diagnosis and are offered on an advisory basis only, in conjunction with the physician's knowledge of ECG patterns, patient background, clinical history, symptoms, and other diagnostic information.	can be integrated into medical devices. In this case, the medical device manufacturer will identify the indication for use depending on the application of their device. DeepRhythmAI is not for use in life-supporting or sustaining systems or ECG Alarm devices. Interpretation results are not intended to be the sole means of diagnosis. It is offered to physicians and clinicians on an advisory basis only in conjunction with the physician's knowledge of ECG patterns, patient background, clinical history, symptoms and other diagnostic information.	monitors, or cardiac alarm devices, or OTC-use only devices. HeartKey Rhythm analysis results are not intended to be the sole means of diagnosis. It is offered to physicians and clinicians on an advisory basis only in conjunction with the physician's knowledge of ECG patterns, patient background, clinical history, symptoms, and other diagnostic information.		data from the VitalConnect Platform are intended for use by healthcare professionals as an aid to diagnosis and treatment. VitalPatch device can be used on patients with pacemakers that comply with ISO 14117:2012 and ANSI/AAMI PC69:2000 without deviations. Heart rate, electro-cardiography (ECG), heart rate variability, R-R interval, and respiratory rate are not intended for patients with pacemakers. The device is not intended for use on critical care patients.
Intended Use	Continuous and automatic analysis of arrhythmias from ECG data and RR interval for use in outpatient cardiac telemetry and patient monitoring in non-critical healthcare settings for non-urgent clinical decision-making	Continuous and automatic analysis of ECG data in adult patients for use in outpatient cardiac telemetry, Holter, event recorder or other similar devices	Continuous and automatic analysis of beats and arrhythmias from ECG data for use in non-active patient monitoring or for non-urgent clinical decision making	Vista Solution Platform is a wireless remote monitoring system intended for display of physiological data and triggering notifications.	The VitalPatch biosensor is intended for the continuous collection of physiological data in home and healthcare settings.
Cardiac Arrhythmia Detection	Atrial fibrillation/atrial flutter	Atrial Fibrillation detection	- Atrial fibrillation - Normal sinus rhythm	Vista Solution does not detect cardiac	Not Applicable

	• AV Block (2nd degree, Type I and II) • Pause • Paroxysmal supraventricular tachycardia (PSVT) • Ventricular tachycardia/run • Sinus bradycardia • Sinus tachycardia • Normal sinus rhythm • Others (inconclusive)	• Non-paced arrhythmias interpretation • Non-paced ventricular arrhythmia calls	• Cardiac Pause • Tachycardia • Bradycardia • Inconclusive • Unreadable	arrhythmias; however, it can display physiological data collected by the VitalConnect platform	
Beat detection	VitalRhythm does not detect beats, however, beat detection such as QRS detection, RR interval, Heart Rate and Heart Rate Variability are provided by the VitalPatch reference device	• QRS detection • Heart rate determination for non-paced adult • Ventricular ectopic beats detection • Supraventricular ectopic beats detection • R-R interval detection	• Beat Clustering - Beat cluster template - Beat cluster count - QRS location - Beat classification • Heart Rate • Beat Analysis - Normal (N) - Ventricular (V) ectopic	Vista Solution does not detect beats, however, it displays beat detection features such as QRS detection, RR interval, Heart Rate and Heart Rate Variability are provided by the VitalPatch reference device	• QRS detection • RR interval • Heart Rate • Heart Rate Variability
Patient Population	Patients who are 18 years of age or older	Adults	Adults over the age of 18	Patients who are 18 years of age or older	Patients who are 18 years of age or older
510(k) Number	K242129	K241197	K233755	K193343	K192757
Device Classification	Class II	Class II	Class II	Class II	Class II
Classification Product Codes	DQK, DPS, QYX	DQK, DPS, QYX	DQK, DPS	DRG	DRG, DSI, MHX
Classification Names	Programmable Diagnostic Computer, Electrocardiograph, Outpatient Cardiac Telemetry	Programmable Diagnostic Computer, Electrocardiograph, Outpatient Cardiac Telemetry	Programmable Diagnostic Computer, Electrocardiograph		
Regulation Numbers	870.1425 870.2340 870.1025	870.1425 870.2340 870.1025	870.1425 870.2340	870.2910	870.2910 870.1025

Use Environment	Home or non-critical healthcare settings	Unknown	Hospital or home	Home or non-critical healthcare settings	Home or non-critical healthcare settings
Components	Software only (SaMD): 1) A web API 2) A cloud-based, automated proprietary ECG interpretation support algorithm	Software only (SaMD): 1) A web API 2) An automated proprietary algorithm	Software only (SaMD): 1) A web API 2) An automated proprietary algorithm	VitalPatch biosensor, the Relay Device (VistaPhone or VistaTablet) preloaded with the VistaPoint software application, the VC Cloud, and the cloud-based VistaCenter user interface	No additional components
Programming Interface	Web application programming interface (API)	Web application programming interface (API)	Web application programming interface (API)	Web application programming interface (API)	VitalConnect Application Programming Interface (API)
Part responsible for ECG signal arrhythmia analysis	The automated proprietary algorithm developed using deep-learning techniques, which measures and analyzes ECG data to provide certified cardiac technicians and qualified healthcare professionals the ability to review and confirm the analytic result	The automated proprietary deep-learning algorithm, which measures and analyzes ECG data to provide qualified healthcare professional with supportive information for review.	The automated proprietary machine-learning algorithm, which measures and analyzes ECG data to provide qualified healthcare professional with supportive information for review	Not Applicable	Not Applicable
Number of leads for a received ECG signal	Single-lead ECG data with sampling frequency of 125 Hz and dynamic range of -10 mV to +10mV	Two-lead ECG data	A single lead ECG data	Display of single-lead ECG data from the VitalPatch reference device	Single-lead ECG data with sampling frequency of 125 Hz and dynamic range of -10 mV to +10mV
Display or Graphical User Interface (GUI)	No primary display or GUI	No primary display or GUI	No primary display or GUI	The VistaCenter user interface is a web-based application that is the primary display for access by	No primary display or GUI

				qualified healthcare professionals.	
Fundamental Scientific Technology	The VitalRhythm application: • accepts ECG and RR-interval data from the VitalPatch device via secure API (Application Programming Interface). • analyzes the data using a proprietary algorithm, which detects specified cardiac arrhythmias. • arrhythmias detected by the software algorithm are returned with interpretation results to be reviewed by a qualified healthcare professional.	The DeepRhythmAI application program interface (API) is intended to be integrated into other device software to provide the following capabilities: • downloads and analyzes data recorded in compatible formats from dedicated ambulatory ECG devices such as Holter, event recorder, Outpatient Cardiac Telemetry devices or other similar recorders. • an automated proprietary deep-learning algorithm, which measures and analyzes ECG data. • a scalable Application Programming Interface (API) to enable easy integration with other medical products.	The HeartKey Rhythm application program interface (API) is intended to be integrated into other device software to provide the following capabilities: • ECG signal conditioning, • ECG signal quality assessment, • QRS beat detection, • Ventricular ectopic beat detection, • ECG rhythm analysis	The Vista Solution Platform: • VitalPatch — a wireless, battery-operated wearable biosensor, worn on the torso to continuously gather physiological data from the person being monitored and then transmits encrypted data via bi-directional Bluetooth communication to the Relay device. • VistaPhone or VistaTablet Relay Device — a phone-based or tablet-based device that runs the VistaPoint application. • VistaPoint — Application that collects, displays and uploads VitalPatch data. • VC Cloud — a secure, cloud-based environment that stores vital sign data and reports. • VistaCenter — a web-based application for monitoring vital sign data from multiple	The VitalPatch device is a wireless, battery-operated wearable biosensor, worn on the torso to continuously gather physiological data from the person being monitored and then transmits encrypted data via bi-directional Bluetooth communication to the Relay device

				patients that runs on a desktop/laptop computer or on a mobile device.	
Continuous Analysis of ECG signals	YES	YES	YES	Display of continuous ECG signal	Collection of continuous ECG signal

Indications for Use Comparison

The VitalRhythm, DeepRhythmAI and HeartKey Rhythm devices are intended for continuous analysis of ECG signals for arrhythmia detection. The indications of the VitalRhythm subject device are similar, but not identical, to the cleared predicate device.

The VitalRhythm subject device, DeepRhythmAI predicate device and HeartKey Rhythm reference device are indicated for analyzing ECG signals and providing the analytical results to healthcare professionals as supportive information. Like the DeepRhythmAI predicate device and HeartKey Rhythm reference device, the VitalRhythm device is not intended to be used with patients requiring life-supporting or life-sustaining systems and the arrhythmia analyses are not intended to be the sole means of diagnosis. Both the VitalRhythm subject device and DeepRhythmAI predicate device continuously and automatically analyze ECG signals for outpatient cardiac telemetry. Both the VitalRhythm subject device and HeartKey Rhythm reference device continuously and automatically analyze ECG signals in either the home or non-critical healthcare settings for non-urgent clinical decision-making.

While the VitalRhythm subject device is compatible with only the VitalConnect reference devices (VitalPatch biosensor and Vista Solution), the DeepRhythmAI and HeartKey Rhythm devices can be integrated into other devices. While the VitalRhythm subject device detects only arrhythmias, the DeepRhythmAI and HeartKey Rhythm devices detect both arrhythmias and beats (QRS detection and ectopic beats). However, beat detection features such as QRS detection, RR interval, Heart Rate and Heart Rate Variability are provided by the compatible, previously cleared VitalConnect reference devices (VitalPatch biosensor and Vista Solution). These differences do not result in a new intended use for VitalRhythm as compared to its predicate devices.

Technological Comparison

The VitalRhythm device's technological characteristics are similar to those of the cleared predicate device.

The VitalRhythm device, DeepRhythmAI device and HeartKey Rhythm device are stand-alone software applications that are compatible with other devices. The VitalRhythm device, DeepRhythmAI device and HeartKey Rhythm device accept continuous ECG signal data (as opposed to retrospective data) from the compatible devices for analysis. The

VitalRhythm device, DeepRhythmAI device and HeartKey Rhythm device use proprietary algorithms based on machine-learning techniques to analyze the ECG signals. VitalRhythm detects cardiac rhythms that are similar to those detected by the DeepRhythmAI predicate device and HeartKey Rhythm reference device.

Both VitalRhythm and the HeartKey Rhythm reference device analyze single-lead ECG data while the DeepRhythmAI predicate device analyzes two-lead ECG data. While the VitalRhythm subject device is compatible with only the VitalConnect reference devices (VitalPatch biosensor and Vista Solution), the DeepRhythmAI predicate device and HeartKey Rhythm reference device can be integrated into other compatible devices. Unlike the DeepRhythmAI predicate device and HeartKey Rhythm reference device, the VitalRhythm subject device does not detect beats; however, beat detection features such as QRS detection, RR interval, Heart Rate and Heart Rate Variability are provided by the compatible, previously cleared VitalConnect reference devices (VitalPatch biosensor and Vista Solution).

In conclusion, the technological differences between VitalRhythm, when used with compatible VitalConnect reference devices, and the DeepRhythmAI predicate device, do not raise new questions of safety or effectiveness. Therefore, the VitalRhythm subject device is substantially equivalent to the predicate device.

VII. Non-Clinical and/or Clinical Tests Summary & Conclusions

Nonclinical testing included de-identified ECG and RR-interval data obtained retrospectively from patients prescribed the VitalPatch biosensor across clinical sites in the United States (US).

The training database was from 354 US clinical sites, and consists of 23,587 patients and 81,391 datasets. The independent test database contains VitalPatch data from 3,309 patients (18 years of age or older) across 7 US testing sites and consists of 7,553 datasets with a total of 22,034 annotated arrhythmia episodes indicating a wide sampling of arrhythmias to support a robust generalization of the algorithm. To ensure an unbiased performance assessment, test datasets are sequestered at the clinical site level.

The performance of VitalRhythm, when assessed using the independent test database, in terms of sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), and accuracy is provided in the table below:

Rhythm	Sample size		Observed performance				
	Patient	Dataset	Sensitivity	Specificity	PPV	NPV	Accuracy
Atrial fibrillation/ flutter	421	1512	99.9	99.9	99.8	99.9	99.9
AV block (2nd degree, Type I and II)	118	716	98.7	100.0	99.9	99.9	99.9
Paroxysmal supraventricular tachycardia	689	1003	98.6	99.9	99.4	99.9	99.9
Ventricular tachycardia/run	610	737	99.3	99.9	99.4	99.9	99.9
Pause	200	960	99.7	100.0	99.9	100.0	100.0
Others (inconclusive)	1547	1923	98.9	99.9	99.2	99.9	99.9
Sinus	3009	5906	99.8	99.9	99.8	99.9	99.8
Normal sinus rhythm	2394	4201	98.9	99.1	99.1	98.9	99.0
Sinus bradycardia	488	1009	97.0	99.8	98.9	99.6	99.5
Sinus tachycardia	914	1161	99.4	99.5	97.7	99.8	99.5

Subgroup analyses performed to evaluate potential heterogeneity (Mann Whitney Wilcoxon test, $p \geq 0.01$) in the detection performance of the VitalRhythm algorithm across demographic factors (patients' age, gender, race, ethnicity and body mass index), dataset durations, and testing sites did not result in statistically significant differences.

The test results demonstrated that performance of the VitalRhythm algorithm, when assessed on the independent test database, in terms of sensitivity, specificity, Positive Predictive Value (PPV), Negative Predictive Value (NPV), and accuracy met the acceptance criteria of $\geq 95\%$ for non-sinus rhythms including Atrial fibrillation/ flutter, AV block (2nd degree, Type I and II), Paroxysmal supraventricular tachycardia, Ventricular tachycardia/run, Pause and Others (Inconclusive). Additionally, the test results showed that sinus rhythms including Normal sinus rhythm, Sinus bradycardia and Sinus tachycardia met the acceptance criteria of $\geq 90\%$ with regard to sensitivity, specificity, PPV, NPV, and accuracy.

No additional clinical testing was required to support this submission.

Substantial Equivalence Conclusion

The nonclinical performance data collected in support of the proposed indication demonstrate that use of VitalRhythm is as safe and as effective and performs as well as or better than the predicate devices to detect target arrhythmias. The scientifically valid performance data establish substantial equivalence and demonstrate that the differences between VitalRhythm and its predicate do not raise different questions of safety and effectiveness.